Methods & Modeling

Day 4 — Building toward a defined target

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Where we are on the spine

Handoff — participants recap their Day 3 evaluation plan.

- **Yesterday:** data → a credible definition of success.
- **Today:** a definition of success → a model **designed to reach it.**

Day 3: estimand & metrics

Method selection is a choice, not a default

- The method should **follow from** the data (Day 2) and the metrics (Day 3).
- The failure mode to avoid: **reverse-justifying a favored technique** — picking the model first, then arguing for it.
- "I want to use a transformer" is not a method rationale.

The honest order

Most projects run this backward.

The temptation

method → problem → data

The discipline

data → target → **method last**

FIG F4.1 **Method comes last.** A pipeline with the arrow drawn deliberately right-to-left for the "temptation" and left-to-right for the "discipline" — same boxes (data, target, method), opposite order. *See build guide F4.1.*

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THE SPACE OF OPTIONS

The method landscape

From regression to foundation models

A **spectrum** of capacity and assumptions — not a ranking:

- **Regression** — interpretable, strong baseline, hard to beat on small tabular data.
- **Gradient-boosted trees** — the workhorse for structured clinical data.
- **Deep nets** — sequences, images, signals; needs data and tuning.
- **Foundation models** — representation reuse; the newest option on the menu.

FIG F4.2 **Capacity vs. assumptions.** A horizontal spectrum from regression → GBTs → deep nets → foundation models, with two axes annotated: rising capacity / data hunger, and what each assumes about the data. *See build guide F4.2.*

Health-specific inductive biases

Health data is **not** generic tabular or text data:

- **Temporality** — events have order and timing that carry meaning.
- **Irregular sampling** — measurements arrive when care happened, not on a grid.
- **Missingness** — *informative*: that a value is missing is a signal.
- **Multimodality** — notes, labs, waveforms, images describe one patient.

FIG F4.3 **What health data demands.** A patient timeline showing irregular, multimodal events with visible gaps — annotated to show order, spacing, missingness, and modality all carrying signal. *See build guide F4.3.*

Worked example — informative missingness

Why "impute the mean and move on" is a modeling decision, not a cleanup step:

- A lab is **missing because it wasn't ordered** — and it wasn't ordered because the clinician wasn't worried.
- The **pattern of measurement** is itself a severity signal.
- Throwing away the missingness mask can discard the most predictive feature you have — or smuggle in leakage.

Day 2: care-process confounding

Day 3: leakage

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THE CENTRAL 2026 DECISION

Foundation models in health AI

Foundation vs. task-specific — the core decision

Foundation model

- Broad pre-training, reused representations.
- Data efficiency downstream; a head start.
- Less control; inherits pre-training biases.

Task-specific model

- Built for one target.
- Wins when the task is narrow, data plentiful, control/interpretability matters.
- No representation to reuse.

Frame it as a **decision**, not an endorsement.

What foundation models for health *are*

- High-capacity models trained over **EHR / clinical data** — learning reusable representations of patients.
- **EHR foundation models** (event streams), **clinical language models**, **multimodal models.**
- What they learn: a transferable representation. What they don't: a finished answer.

FIG F4.4 **Pre-train then adapt.** Two-stage schematic: large unlabeled clinical data → pre-trained representation → (small labeled task) → adapted model. Three downstream tasks branching off one representation. *See build guide F4.4.*

What they are *not*

- **Not** a general clinical intelligence.
- **Not** exempt from the evaluation rigor of Day 3 — a foundation model still has to clear every bar.
- **Not** automatically fair, calibrated, or transportable because it is large.

> Scale buys a representation. It does not buy a contribution.

↩ Day 3: every model clears the same bar

Do we still need clinical language models?

The honest counterpoint (Lehman et al., 2023; Agrawal et al., 2022):

- A **general** LLM with good prompting can match specialized clinical models on some tasks — **few-shot clinical extraction** is real.
- ...and a smaller **specialized** model still wins on others (cost, control, latency, privacy).
- The answer is **task-dependent** — which is exactly the point.

Fine-tune, prompt, or train from scratch?

A decision keyed to **data volume, task, and resources:**

- **Train from scratch** — large in-domain data, a task no pre-trained model fits.
- **Fine-tune** — a relevant pre-trained model exists; moderate labeled data.
- **Prompt / in-context** — little labeled data; fast iteration; accept less control.

FIG F4.5 **A decision tree.** Branching on: relevant pre-trained model? → labeled data volume? → control/cost needs? → leaf = scratch / fine-tune / prompt. Make it something a participant can trace for their own project. *See build guide F4.5.*

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Representation learning — and where it bites

- A good **representation** is what makes downstream health tasks tractable.
- But representation choices **propagate**: a representation that encodes a shortcut hands that shortcut to every task built on it.
- **Copycats** (Jimenez-Sanchez et al., 2024): a public dataset's quirks get baked in and re-used field-wide.

↩ Day 3: shortcut learning

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BUILDING WELL

Baselines, structure, and trust

Data structure constrains method

- Data **standards** — e.g. **MEDS** — shape what a model can consume.
- Method choice is **entangled** with data representation: you cannot choose a method independently of how the data is structured.
- The data spec you wrote on **Day 2** already narrowed today's method menu.

FIG F4.6 **Representation sets the menu.** A MEDS event stream feeding into the subset of methods that can consume it — the same stream gating which models on the Slide 6 spectrum are even available. *See build guide F4.6.*

Day 2: MEDS

Baselines first; internal validation

- Check the model **while building** — not only at the end.
- **Baselines first.** A complex model that cannot beat regression has not earned its complexity.
- This is **internal validity** — the model on its own terms. The Day 4 rung of the validity ladder.

Day 3: the validity ladder

The afternoon — a fair three-way comparison

The 2:45 notebook, head-to-head on one task:

Logistic regression
the baseline bar

Gradient-boosted trees
the structured-data workhorse

Neural network
capacity, if the data earns it

Same cohort, same metrics (Day 3), same split — so the comparison is honest.

Method, data, metric are entangled

- The three cannot be chosen independently — each constrains the others.
- **Trustworthy-by-construction:** build so failures don't hide, rather than auditing them in afterward.
- Tomorrow's question — does it survive a clinic? — starts in how you build today.

→ Day 5: deployment

Propose, then defend

Today's deliverable — Method rationale + 1-slide proposal.
Your **approach**, justified against your **data** (Day 2) and your **metrics** (Day 3) — with a credible **alternative** named.

Now (9:00): draft your 1-slide method proposal.
Next: 11:00 guest (Walter Gerych) — trustworthy-by-construction · 1:30 peer review · 2:45 methods-comparison notebook.